

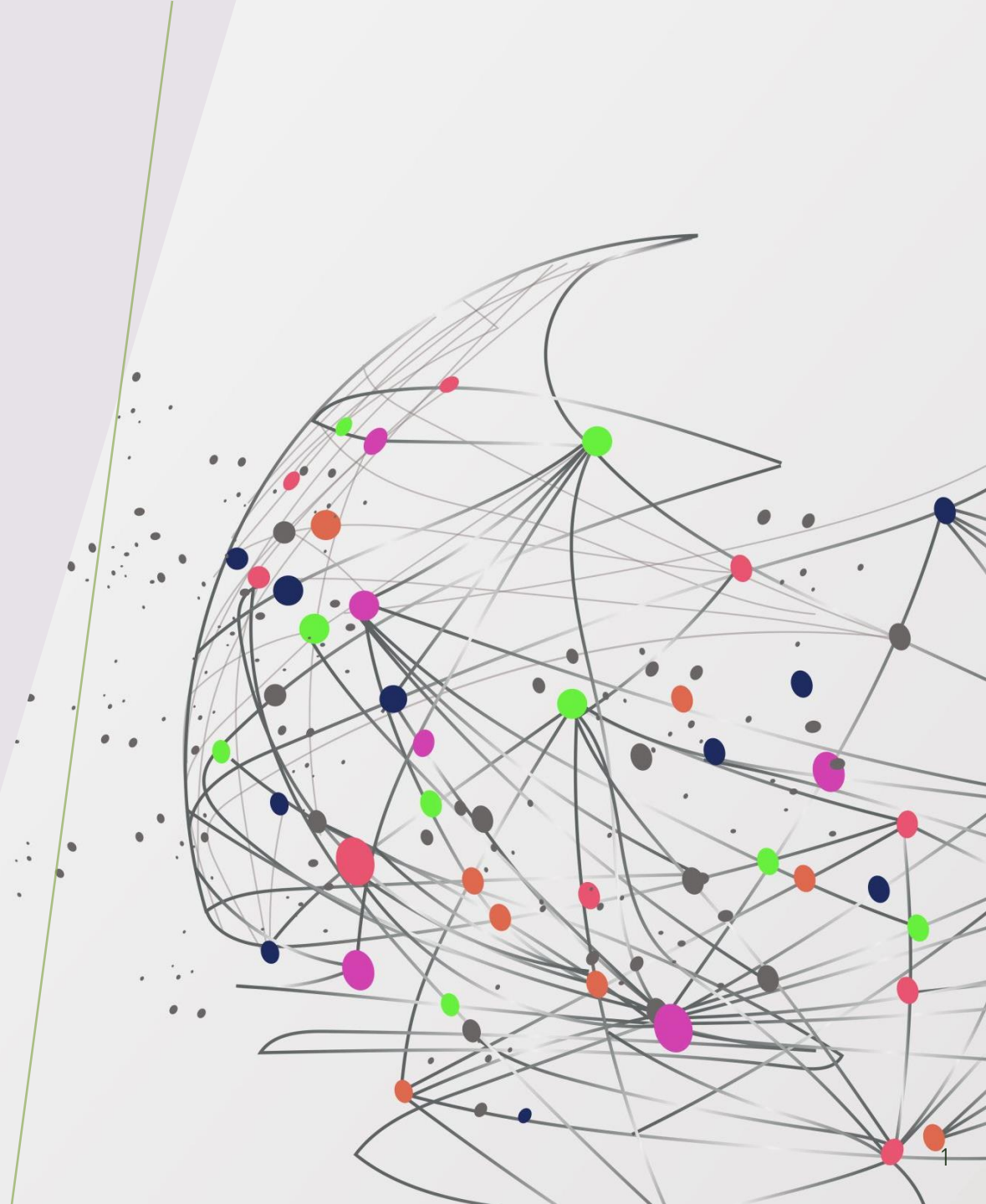
ASIAN WOMEN ADVANCING AI

LEARN & LEAD: RAG

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CHALLENGES WITH TRADITIONAL CHATGPT FOR FILE-BASED QUERIES



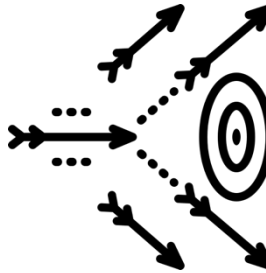
Cutoff Date

ChatGPT lacks access to external data once trained, limiting its knowledge scope to the cutoff date.



Static

Cannot dynamically ingest and process user-provided files during a session.



Accuracy

Struggles with answering questions accurately for large or complex documents without specific context.



Hallucination

May hallucinate information if the requested data is outside its training set.

WHY RAG IS THE SOLUTION?

Real-Time Data Access

Combines generative AI with up-to-date information from external sources.

File Integration

Dynamically retrieves and processes information from large, complex documents.

Improved Accuracy

Provides grounded, evidence-backed answers by retrieving relevant context.

Scalability

Efficiently handles large datasets or documents without overwhelming memory constraints.

RAG ARCHITECTURE OVERVIEW

RAG enhances LLM responses by combining them with relevant information retrieved from a knowledge base, leading to more accurate and contextual outputs.

1. Query Processing

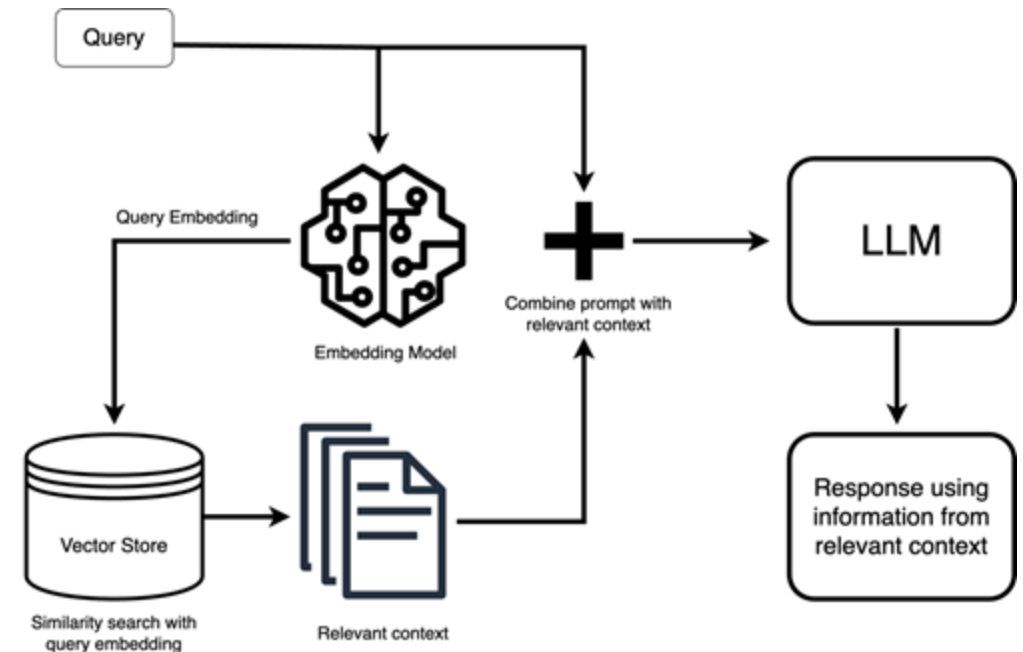
- User query is converted to vector embedding
- Same embedding model processes both query and documents

2. Information Retrieval

- Query embedding used for similarity search in vector store
- Most relevant documents are retrieved as context

3. Response Generation

- Query and retrieved context combined into prompt
- LLM generates response using both prompt and context
- Final output combines retrieved information with LLM's knowledge



[Source](#)

WHY RAG STANDS OUT?

Personalization

Tailor responses based on user context and retrieved data.

Dynamic Updates

Works with live databases, ensuring fresh and relevant information.

Cost Efficiency

Reduces computation by focusing only on relevant parts of data. No need to train a Large Language Model.

Modularity

Flexibility to integrate with existing systems like CRMs, analytics tools, etc.

Data Privacy

Enables secure queries on private datasets, protecting sensitive information from external exposure.

LIMITATIONS OF RAG

Data Quality Dependency

The system's output is only as good as its knowledge base - poor, outdated, or missing data leads to inaccurate or incomplete responses.

Computational Overload

Requires significant computational resources for both retrieval and generation, causing increased latency and higher operational costs

Retrieval Accuracy

The retrieval component may fail to fetch the most relevant information or misinterpret semantic relationships, leading to irrelevant responses

Context Integration

Systems struggle to seamlessly blend retrieved information with generated content, often producing disjointed or incoherent responses

and some more...

RAG IN ACTION



Customer Support

Instant and accurate responses from support documentation or FAQs.



Healthcare

Medical diagnosis support by retrieving relevant studies and patient records.



Education

Personalized tutoring by fetching information from course materials.



Finance

Extracting insights from market reports, financial data, and client queries.

[More Use Cases of RAG](#)

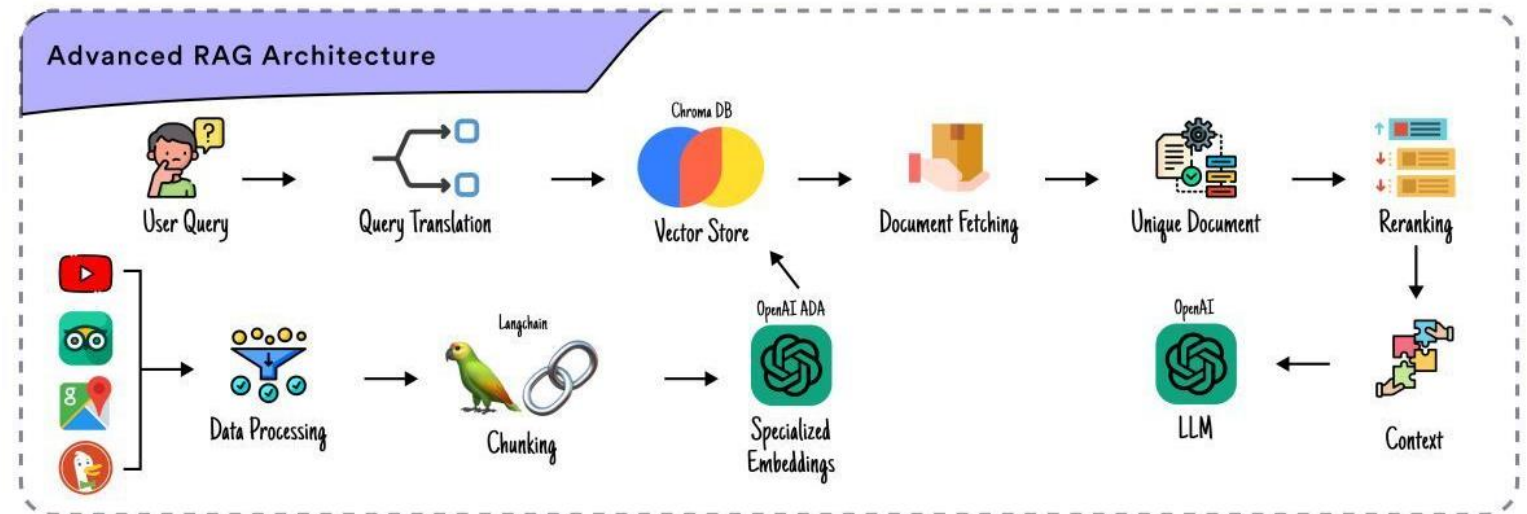
INTRIPID

Intripid, is an AI-powered travel planner. Based on your personality and your preferences it creates a travel itinerary for you for your trip.

What's Next? is a website where you put your location and your preferences, and it will suggest you a place to go next.

What's Next?

<https://dev.intripid.co/whats-next>
(open on mobile)



*The Present RAG Architecture, supporting our Agents to perform seamless generation.

INDUSTRY LEVEL RAG IMPLEMENTATION EXAMPLE



Sendbird AI chatbot

[Shopify Sendbird AI Chatbot using RAG to query Shopify's data](#)



[Morgan Stanley's RAG Architecture for Wealth Management](#)

Read more: <https://www.projectpro.io/article/rag-use-cases-and-applications/1059>

RAMO: Retrieval-Augmented Generation for Enhancing MOOCs Recommendations

×

OpenAI API Key

[Get an OpenAI API key](#)

Coursera Free Courses Recommender 🤖 RAMO

🔥 A streamlit app recommending courses for learners from different backgrounds to get personalized Coursera course recommendations.

This is a explainable recommendation system that recommends Coursera courses based on your needs.

Coursera-rec uses the [OpenAI API](#) and langchain library to recommend courses based on your needs. Some courses' url are expired since the dataset we use is for year 2023, thanks for understanding, we'll update it once we have the latest data

Coursera Courses: Top 5 Courses by Course Rating

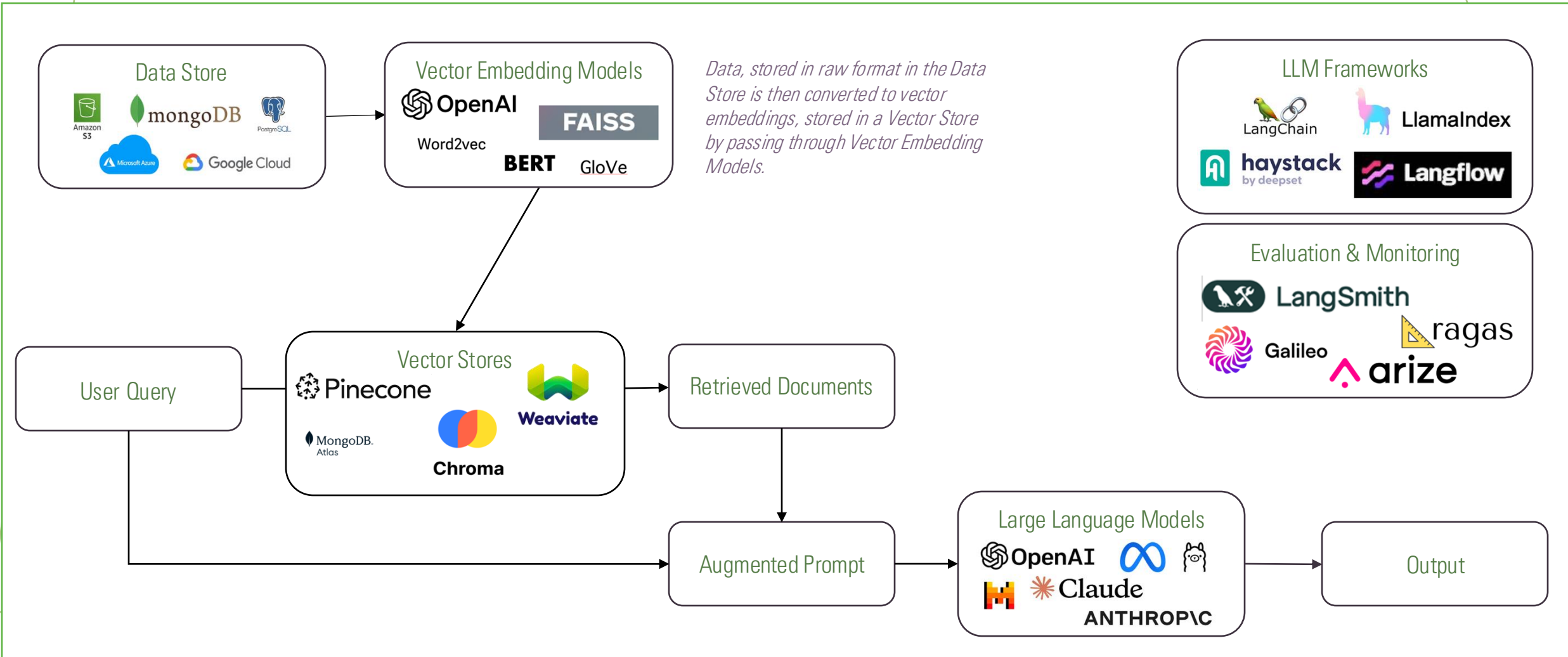
title	course_by	ratings	url
Capacitación sobre COVID-19 para trabajadores de salud	Stanford University	4.9	https://www.coursera.org/learn/covid-19-capacitacion-trabajadores-de-la-salud
Manejo de Crisis Epilépticas y Promoción de la Inclusión Social	Pontificia Universidad Católica de Chile	4.9	https://www.coursera.org/learn/manejopromocion
Advanced Modeling for Discrete Optimization	The University of Melbourne	4.9	https://www.coursera.org/learn/advanced-modeling
Patrick Henry: Forgotten Founder	University of Virginia	4.9	https://www.coursera.org/learn/henry
Exploring Beethoven's Piano Sonatas Part 4	Curtis Institute of Music	4.9	https://www.coursera.org/learn/exploring-beethoven-piano-sonatas-4

Enter text:

Enter your question here...

Submit

TECH STACK FOR RAG



RAG EVALUATION

Context Relevance

Measures how well retrieved documents align with the user's query. High context relevance indicates the system is fetching the most appropriate information needed to answer the question.

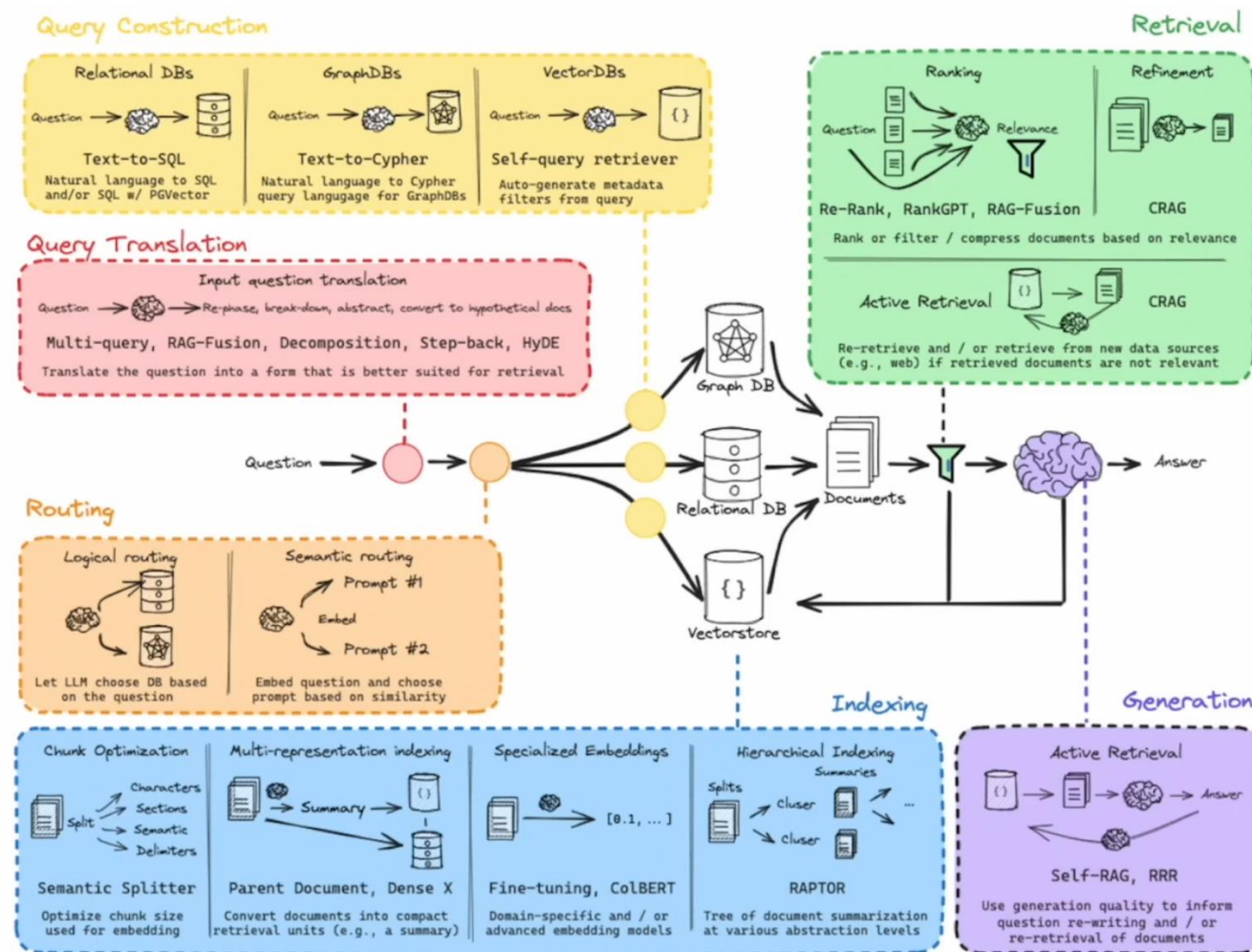
Answer Relevance

Assesses if the final response directly addresses the user's question with appropriate information. It measures both semantic alignment and completeness of the answer relative to the query.

Groundedness/Faithfulness

Evaluates if the generated response is factually supported by the retrieved context, ensuring every statement can be traced back to source documents. This prevents hallucination and maintain factual accuracy.

IMPROVING THE RAG



[Watch this video to understand the above diagram](#)

THAT'S NOT IT...

Corrective RAG

CAAG

Fusion RAG

Speculative RAG

Graph RAG

Self RAG

Agentic RAG

Modular RAG

Radio RAG

Read more on this: <https://www.projectpro.io/article/rag-architecture/1079>

Demo Time!

LET'S BUILD THIS TOGETHER?

- Workshop: 12th February 2025, 8 PM EST
- Pre-requisites:
 - Laptop (obviously 😊)
 - OpenAI API Key – we'll share a document detailing the steps!
 - Google Drive account – any Gmail account would do!
 - Any PDF File you want to ask questions to!
 - Install GitHub (optional – needed only when deploying the chatbot on Streamlit Cloud)
 - GitHub Installation Instructions for Windows 11: <https://www.youtube.com/watch?v=AdzKzlp66sQ>
 - GitHub Installation Instructions for Mac: <https://www.youtube.com/watch?v=p0Js7lF17yl>

REFERENCES & RESOURCES

1. Learn RAG from Scratch: <https://www.youtube.com/watch?v=sVcwVQRHlc8>
2. Top 7 RAG Use Cases and Applications to Explore in 2025: <https://www.projectpro.io/article/rag-use-cases-and-applications/1059>.
3. What is RAG: <https://www.clarifai.com/blog/what-is-rag-retrieval-augmented-generation>
4. DataCamp Course: RAG using LangChain by Meri Nova
5. Comprehensive Guide to RAG Tech Stack: <https://www.useparagon.com/blog/comprehensive-guide-to-the-rag-tech-stack>
6. What is RAG Architecture? <https://www.projectpro.io/article/rag-architecture/1079>
7. 5-day Gen AI Intensive Course with Google and Kaggle: <https://www.kaggle.com/learn-guide/5-day-genai#GenAI>

THANK YOU!

Any questions?